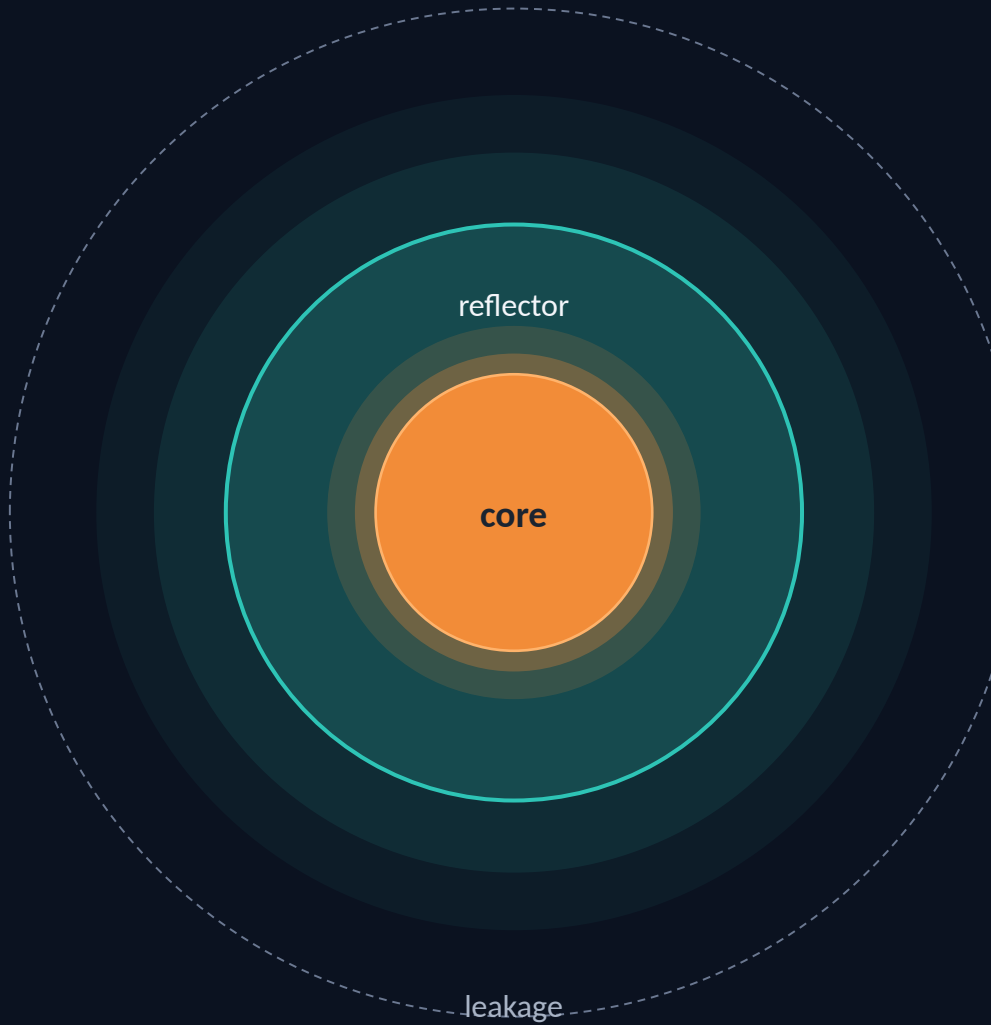
NSEN 6684 · REACTOR PHYSICS · IDAHO STATE UNIVERSITY

Beyond the Six-Factor Formula

Neutron Leakage, Spectral Effects, and Reactivity Feedback in Compact Space Fission Reactors

Thesis: classical life-cycle theory is the foundation for criticality, but compact fast-spectrum space reactors require transport, leakage, reflector and temperature-dependent reactivity analysis.



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ROADMAP — click to jump

Eigenvalue | Diffusion | Leakage | Reflectors | Spectrum | KRUSTY | Findings

Academic coursework · public sources only · calculations use idealized benchmark constants

Why Compact Fast Reactors Are Different

~14 days

lunar night at low latitude

≥ 354 h

darkness Lunar Reactor-1 must ride through (draft, as reported)

11 × 25 cm

KRUSTY HEU core: diameter × height

	Large thermal reactor	Compact fast core
Biggest neutron loss	absorption outside fuel	leakage (45–75 % in this study)
Reactivity control	absorber rods, soluble boron	reflector position
Main prompt feedback	Doppler, moderator density	fuel thermal expansion

Leakage is not a correction factor here. It is the physics.

Sources: NASA & DOE (Jan. 2026); New Space Economy (Sep. 2026), reporting NASA draft LR-1 requirements; Sanchez et al., Nucl. Technol. 206 (2020) S56; this work (study 2).

The Classical Life Cycle in Two Minutes

$$k_{eff} = \eta \cdot \varepsilon \cdot p \cdot f \cdot P_{FNL} \cdot P_{TNL}$$

1000	× 1.84	× 1.03	× 0.80	× 0.78	× 0.95	× 0.90	1011
absorbed in fuel	η	ε	P_FNL	p	P_TNL	f	next generation

Assumed LWR-like factors → k_eff = 1.011

Where it breaks in a fast core

No moderator
p and f lose their meaning

One overlapping spectrum
no separate fast and thermal groups

Leakage dominates
not a correction factor

Use instead: **k = production / (absorption + leakage)**

Sources: Lamarsh & Baratta (2001) ch. 6; Duderstadt & Hamilton (1976). Factor values assumed for review.

Generalized Neutron Balance: the k-Eigenvalue Problem

Transport equation

$\Omega \cdot \nabla \psi + \Sigma_t \psi$ = scattering in + $(1/k) \cdot$fission

Operator form

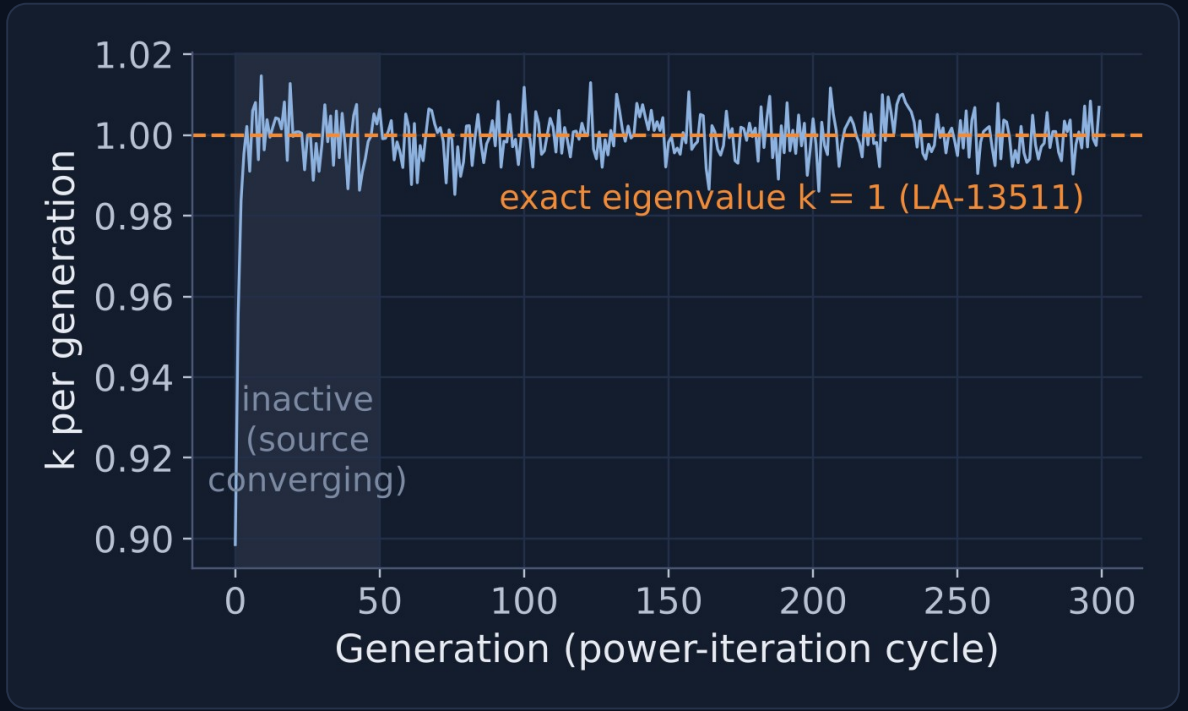
$$L \psi = (1/k) F \psi$$

k_{eff} = dominant eigenvalue

Power iteration

source → transport one generation → new source; k = new / old

Six-factor = two-group, separable special case



My Monte Carlo code: k = 0.9997 ± 0.0003 vs exact 1

Sources: Bell & Glasstone (1970); Duderstadt & Hamilton (1976) ch. 4; benchmark: Sood, Forster & Parsons, LA-13511 (1999), problem Ua-1-0-SP.

From Transport to One-Group Diffusion

1 Balance

$$\nabla\cdot J + \Sigma_a \varphi = (1/k)\, \nu\Sigma_f \varphi$$

integrate over angle and energy

2 Fick's law (P_1)

$$J = -D\,\nabla\varphi, \quad D = 1 / 3\Sigma_{tr}$$

assume linearly anisotropic flux

3 Helmholtz

$$\nabla^2\varphi + B^2\varphi = 0$$

$B_g^2 = (\pi / R_e)^2$, $R_e = R + 0.71\,\lambda_{tr}$

4 Criticality

$$k = k_\infty / (1 + L^2B^2)$$

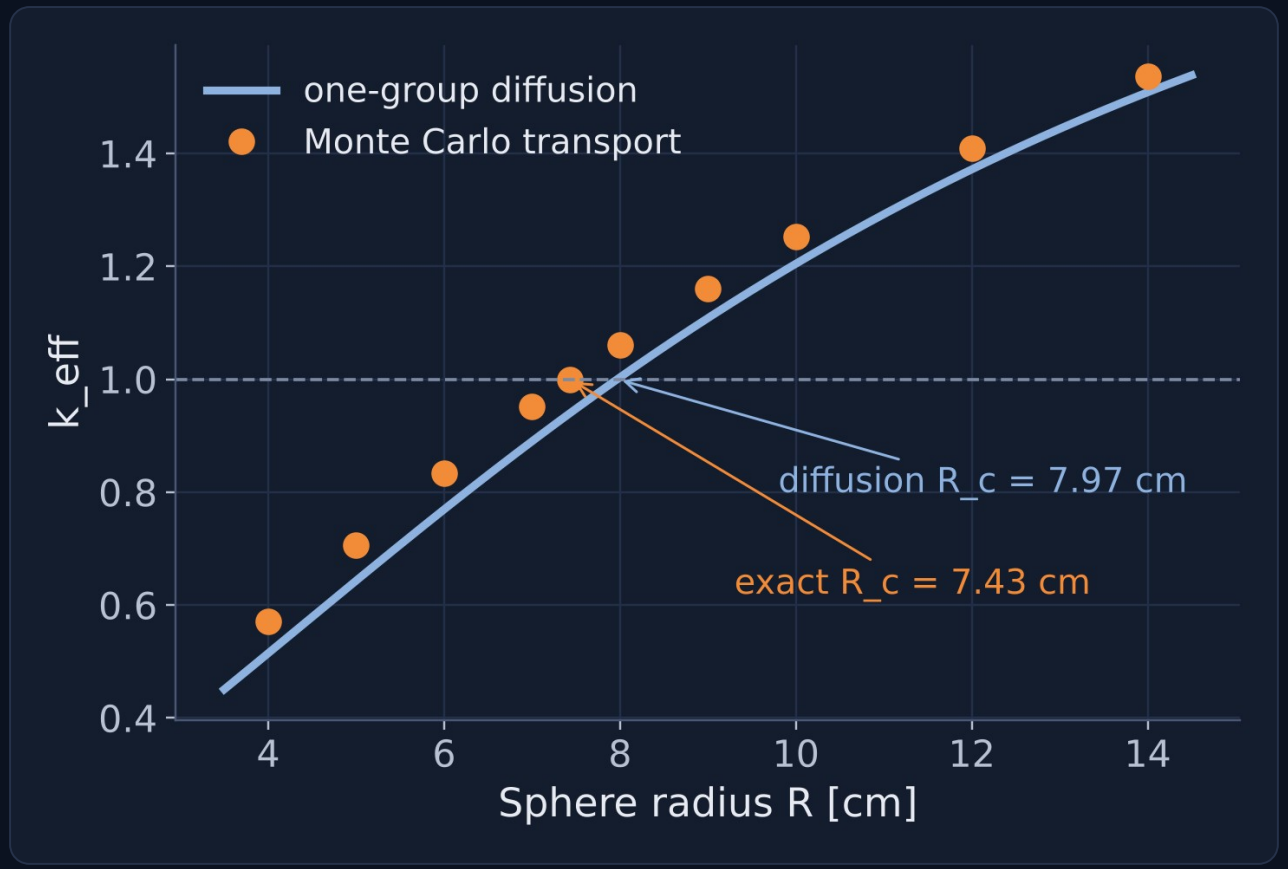
$P_{NL} = 1 / (1 + L^2B^2)$, $L^2 = D / \Sigma_a$

Diffusion is valid when the flux varies slowly over a mean free path · absorption is weak · far from boundaries

A compact fast core fails all three: R is only 2–3 λ_{tr}, and most neutrons see the surface.

Sources: Duderstadt & Hamilton (1976) ch. 4–5; Stacey (2018); Lamarsh & Baratta (2001) ch. 5–6. 0.7104 λ_{tr}: Milne-problem extrapolation distance.

Sensitivity Study: k_eff vs Size and Leakage



R [cm]	k transport	k diffusion	leakage
4.00	0.571	0.515	75 %
6.00	0.834	0.769	63 %
7.43	1.000	0.940	56 %
9.00	1.161	1.108	48 %
12.00	1.409	1.372	37 %
14.00	1.537	1.508	32 %

Diffusion: R_c is 7.3 % too large. k error shrinks from 9.8 % (R = 1.3 λ_tr) to 1.9 % (R = 4.6 λ_tr).

Idealized LA-13511 one-group constants · mathematical illustration, not a reactor calculation

Sources: This work (code/run_analysis.py, study 2); cross sections and exact R_c = 7.428998 cm from Sood et al., LA-13511 (1999).

Reflectors from Boundary Conditions

Core: $\varphi_c = A \sin(Br) / r$

Reflector: $\varphi_r = C \sinh[\kappa(R+T_e-r)] / r$

At r = R: $\varphi_c = \varphi_r, \quad D_c \varphi_c' = D_r \varphi_r'$

Reflector savings $\delta = R_{c,bare} - R_{c,reflected}$

1.0–1.4 cm

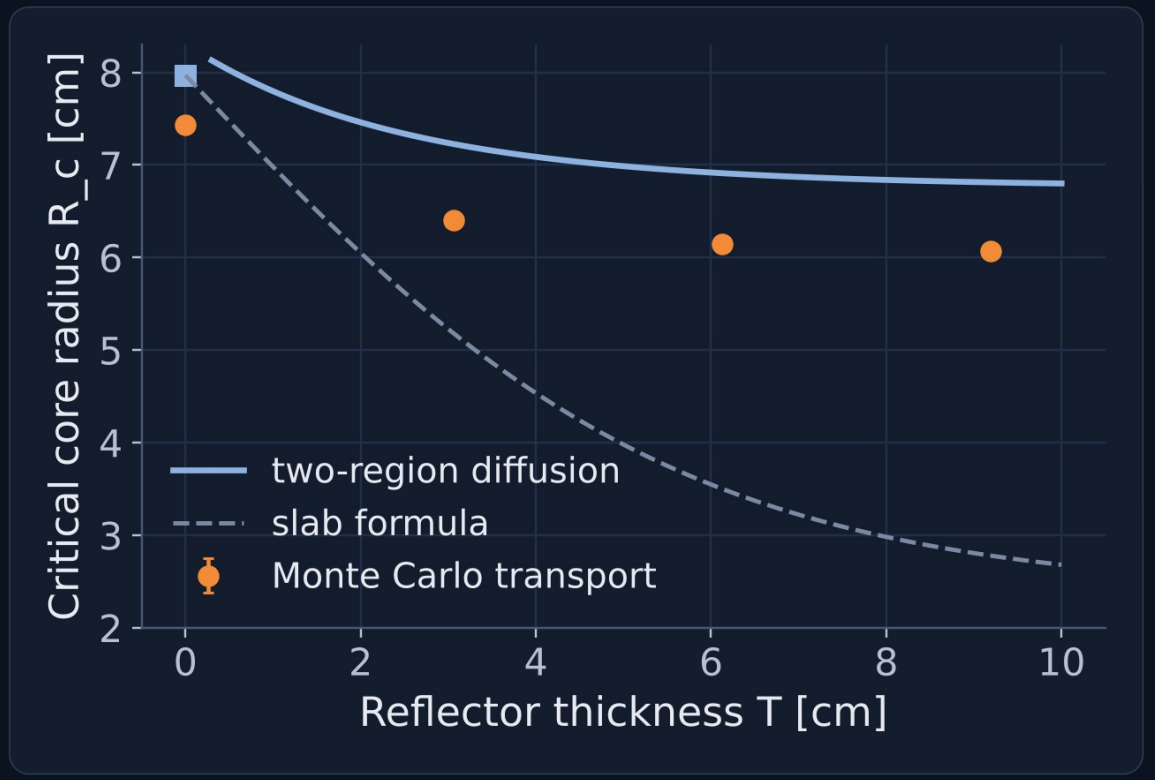
transport savings, 1–3 mfp

×2.7–3.8

slab-formula overestimate

15–19 %

fissions by reflector-returned neutrons

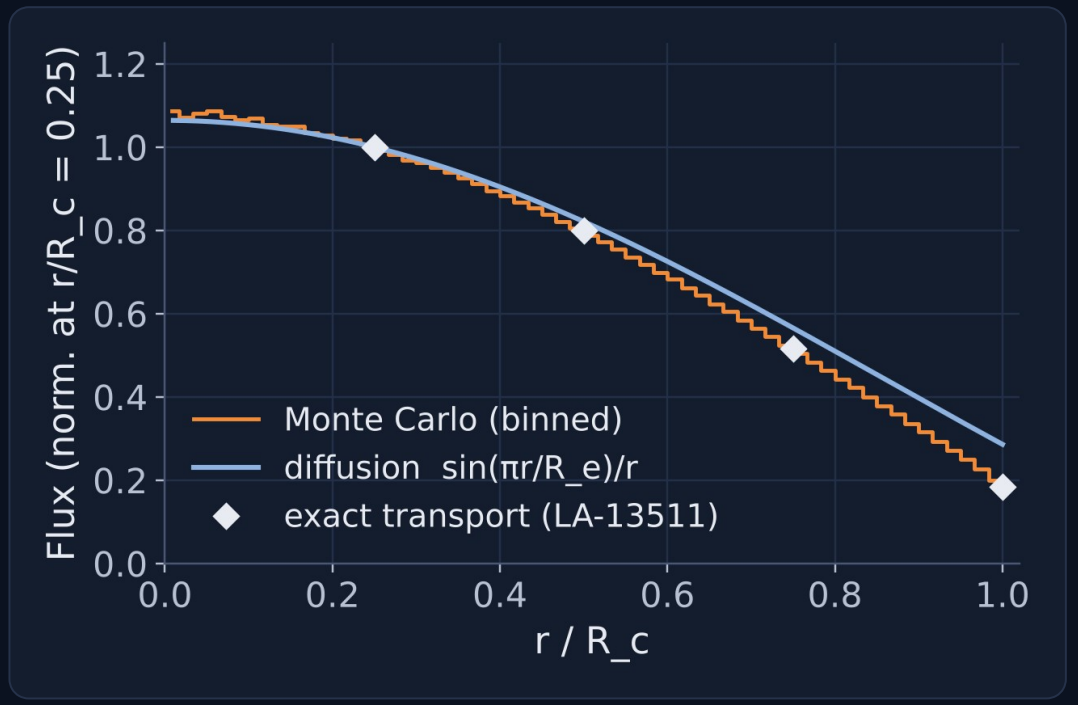


Sources: This work (study 3); H_2O (c = 0.90) and U-235 one-group constants from LA-13511; slab formula: Lamarsh & Baratta (2001) §6.6.

Transport Check: Where Diffusion Fails

LA-13511 problem	exact	Monte Carlo	diffusion
bare sphere	1	0.9997 ± 0.0003	0.940
+1 mfp reflector	1	1.0001 ± 0.0003	0.883
+2 mfp reflector	1	1.0000 ± 0.0003	0.893
+3 mfp reflector	1	1.0001 ± 0.0003	0.897
two-group sphere	1	1.0001 ± 0.0003	0.937

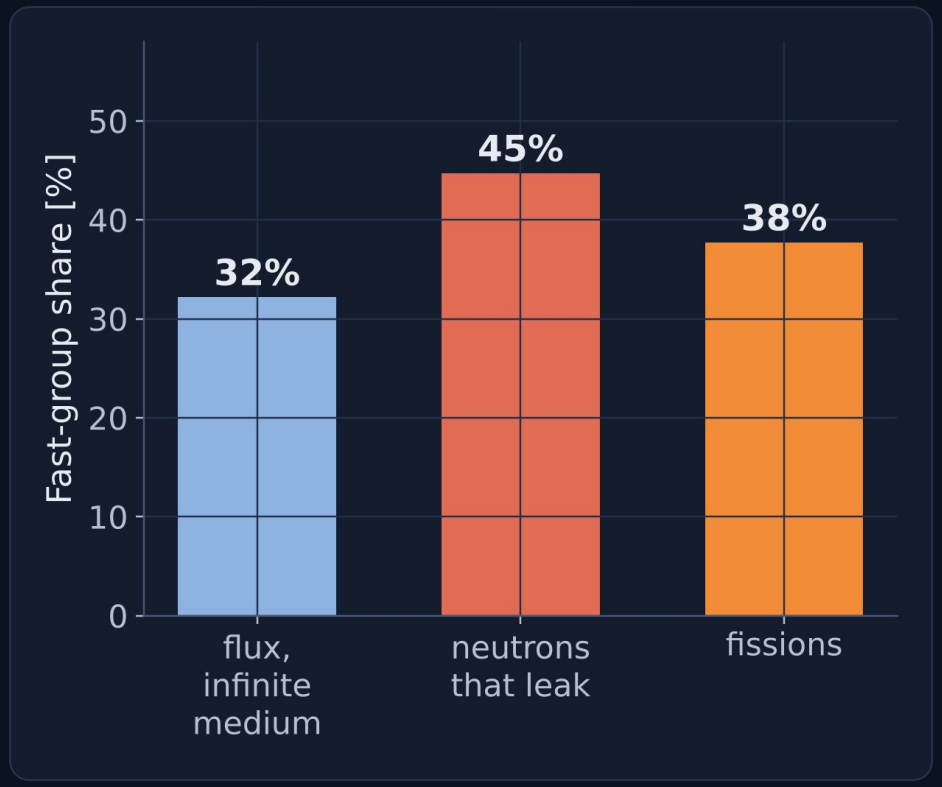
Monte Carlo verified on all five exact solutions (within ≈1σ).
Diffusion is 6–12 % low at the same critical configurations.

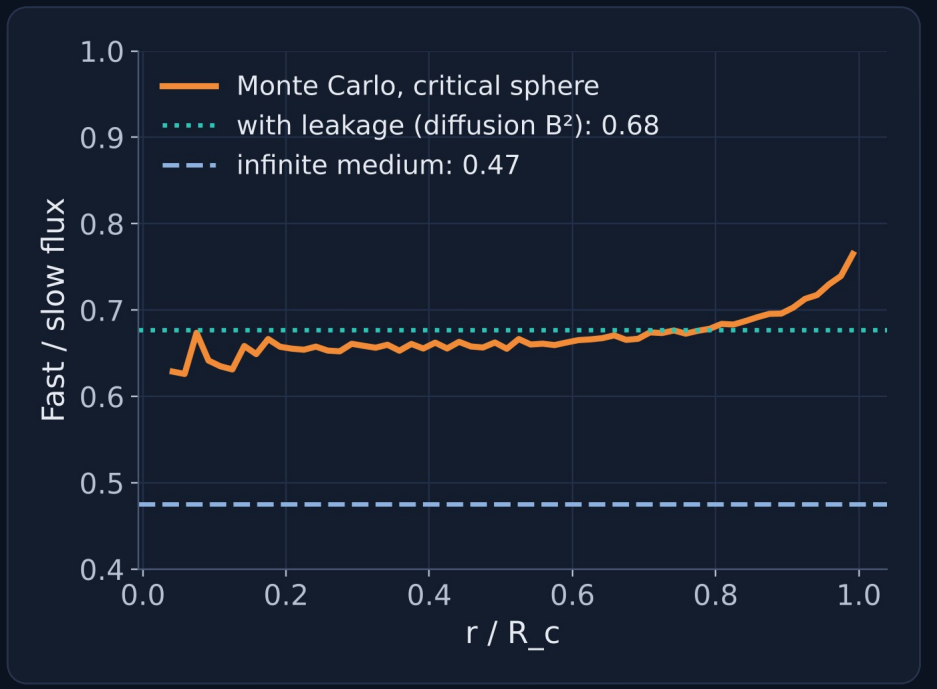


Surface flux: exact 0.18 · MC 0.19 · diffusion 0.29

Sources: This work (studies 1–2); exact solutions and flux ratios: Sood, Forster & Parsons, LA-13511 (1999), Tables 13–15, 35.

Spectral Effects: Leakage Hardens the Spectrum





-1.1 %

R_c error, one-group collapse with infinite-medium spectrum

-0.6 %

with leakage-weighted spectrum

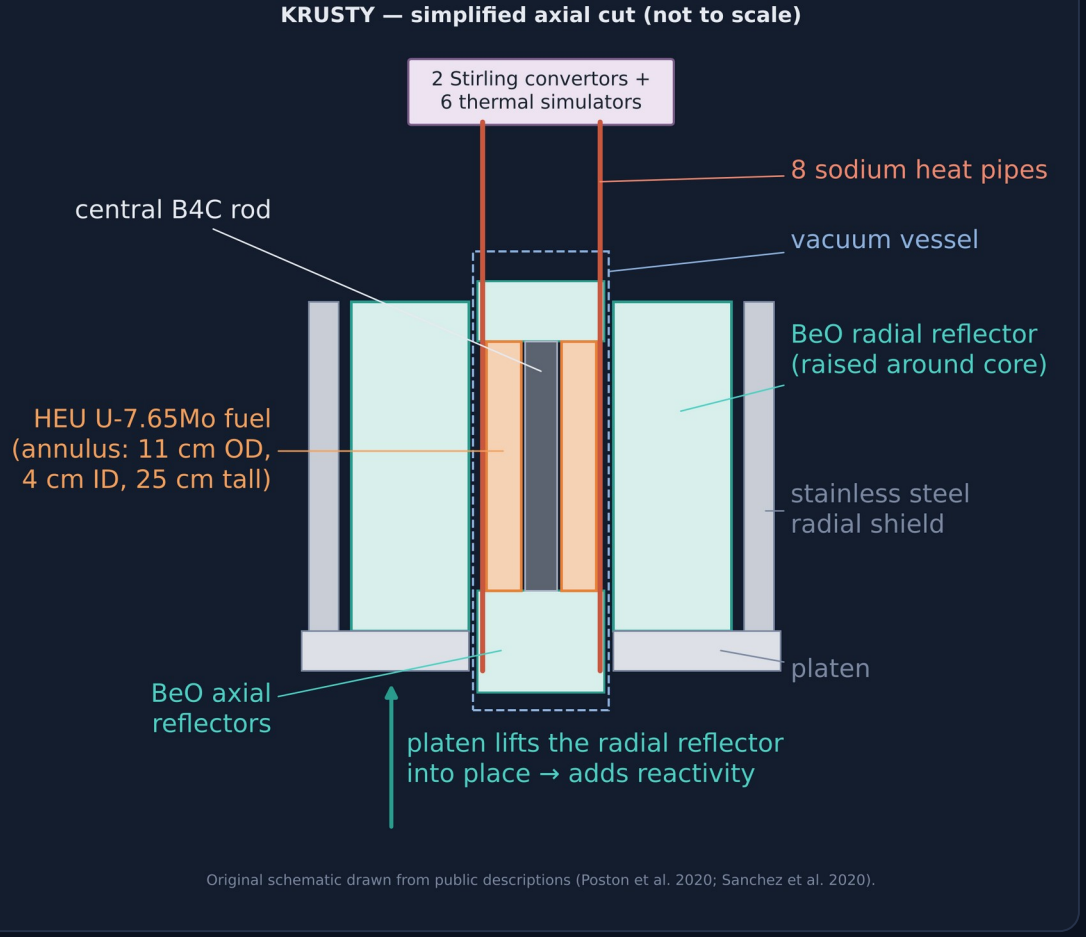
+7.7 %

two-group diffusion error (for scale)

KRUSTY (literature): BeO-moderated neutrons returning from the reflector peak fission at the fuel edge.

Sources: This work (study 4) with LA-13511 two-group U-235 constants (Tables 33–35); Poston et al., KRUSTY Reactor Design, Nucl. Technol. 206 (2020) S13.

KRUSTY: Reflector Worth Is the Control



10.7–12 ¢/mm

platen (reflector) worth, measured

≈ 4 ¢/mm

central B_4C rod — one-third as sensitive

1.42 $

inserted to reach 800 °C

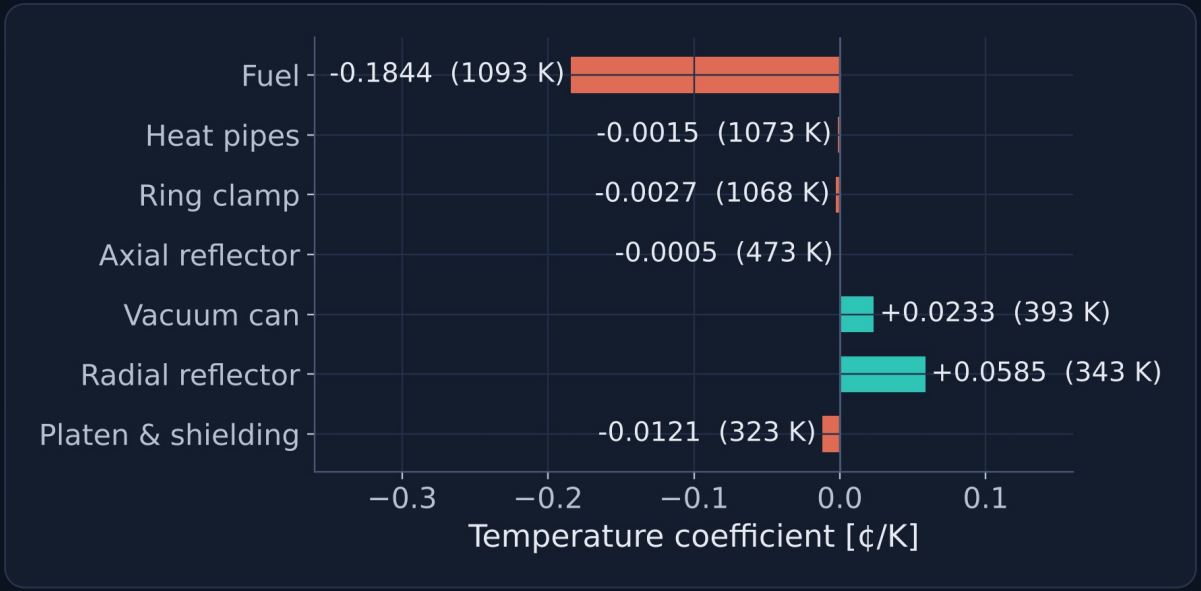
−148 ¢

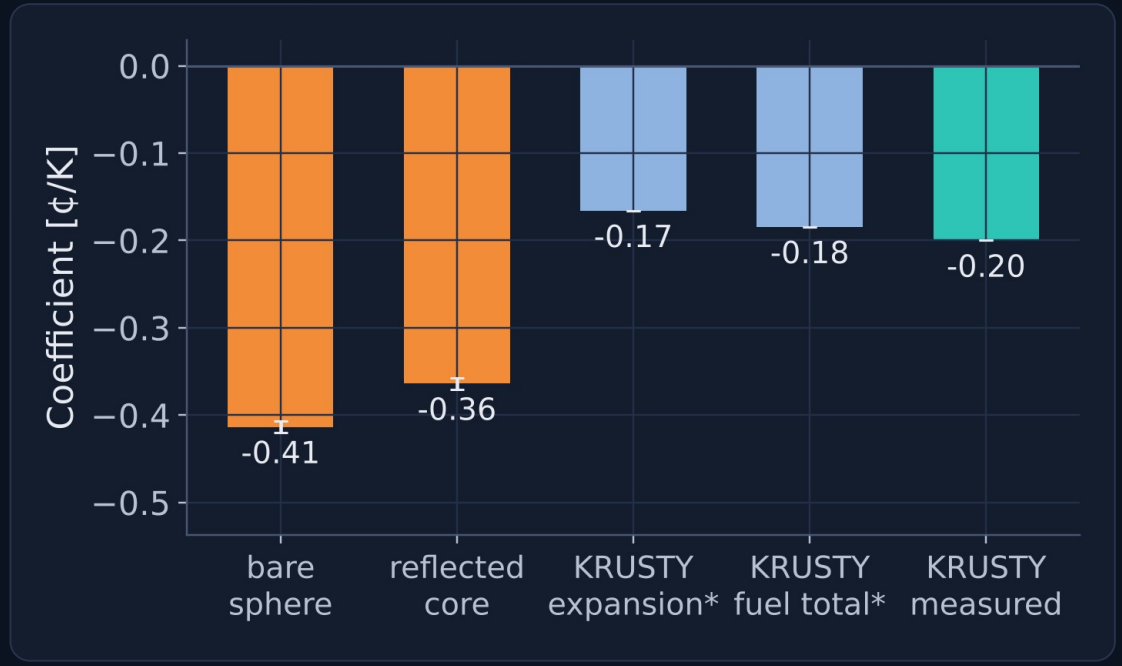
fuel temperature defect (design model)

Inference: the reactivity added by moving the reflector ≈ the reactivity removed by hot fuel. MCNP predicted the first critical within 30 ¢.

Sources: Poston et al., Results of the KRUSTY Nuclear System Test (2020) S89; KRUSTY Reactor Design (2020) S13, Table II; Results of the KRUSTY Warm Critical Experiments (2020) S78.

Decomposing KRUSTY's Negative Feedback





≈ 95 %

of net feedback from fuel heating

90 / 10

expansion / cross sections (incl. Doppler)

+0.06 ¢/K

BeO radial reflector at 343 K

This work: (dln k/dε)·α/β ≈ -0.41 to -0.36 ¢/K. Right sign and order of magnitude; ~2× KRUSTY. A consistency check, not a prediction.

Sources: Poston et al., KRUSTY Reactor Design (2020), Table II and text; Results of the KRUSTY Nuclear System Test (2020); α: Burkes et al., INL/EXT-10-19373 (U-10Mo proxy); this work (study 5).

Findings: Calculated · Observed · Inferred

Calculated (this work)

- Diffusion: k 6–12 % low, R_c +7 % in compact benchmarks
- Error is set by size in mean free paths
- Leakage removes fast neutrons; spectral collapse error ~1 %

Observed (KRUSTY test)

- Reflector position was the reactivity control
- Feedback ≈ 0.2 ¢/°C near 800 °C
- Power followed load passively

Inferred (engineering)

- Compact-core criticality is mostly geometry and leakage
- Feedback sign depends on which component heats
- Design conclusions need validated transport

Next: continuous-energy OpenMC model, validated against the KRUSTY benchmarks (input provided, not yet run)

Sources: This work (studies 1–5); Poston et al. (2020) S89, S13; Stolte et al., Nucl. Technol. 208 (2022) 625; Romano et al., Ann. Nucl. Energy 82 (2015) 90.

Conclusions

1

Six-factor is a special case

of the k-eigenvalue problem: useful for thinking, not for small fast cores.

2

Leakage, reflection and spectrum are coupled

One-group diffusion misses them by 5–12 % in k: many dollars of reactivity.

3

KRUSTY makes it real

Reflector position controlled the reactor; fuel expansion stabilized it.

Classical theory is where the analysis starts. Validated transport is where design conclusions come from.

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References

Slides, report and code: GitHub repository linked in the video description

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